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Patent Public Search | Text View

United States Patent Application Publication

20250285144

Kind Code

A1

Publication Date

September 11, 2025

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SYSTEMS AND METHODS FOR REAL-TIME BIDDING

Abstract

A real-time bidding method includes receiving user input data, generating a first machine learning model that generates a predicted expected performance based on the user input data, and adjusting at least one bid on at least one of at least one keyword and at least one product associated with at least one marketplace, based on the predicted expected performance.

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Family ID: 1000008516000

Appl. No.: 19/073685

Filed: March 07, 2025

Related U.S. Application Data

us-provisional-application US 63563090 20240308

Publication Classification

Int. Cl.: G06Q30/0273 (20230101); G06Q30/0201 (20230101)

U.S. Cl.:

CPC G06Q30/0275 (20130101); G06Q30/0206 (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This U.S. non-provisional patent application claims the benefit of and priority to U.S. provisional patent application Ser. No. 63/563,090, filed Mar. 8, 2024, the entire disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to commerce systems and methods, and more specifically, to systems and methods for real-time bidding.

BACKGROUND

[0003] Commerce systems are well known in the art and are effective means to allow for the transaction of products, commodities, services and the like from one party to another. Commonly, commerce systems are embodied by a market, where many products are offered for sale and people that are customers are able to shop or browse the products and select items for purchase. Such markets may be managed by companies that include Ebay®, Amazon®, Wayfair®, Costco®, Walmart®, and Target®, among others. With the advent of digital marketplaces, sellers are allowed to list products for purchase to anyone with an internet connection. Commonly, many sellers will offer the same or similar products. Shoppers (e.g., users accessing digital marketplaces via the internet) are able to sort through and browse all of these products to find what they are looking for.

SUMMARY OF THE DISCLOSURE

[0004] The various systems and methods of the present disclosure have been developed in response to the present state of the art, and in particular, in response to the problems and needs in the art that have not yet been fully solved by currently available digital marketplaces.

[0005] An aspect of the disclosed embodiments includes a real-time bidding system that includes a computing device, and a memory. The memory includes instructions that, when executed by the computing device, cause the real-time bidding system to: receive user input data; generate at least one model for a target spend value based on the user input data; and adjust at least one bid on at least one of at least one keyword and at least one product, based on output from the at least one model.

[0006] Another aspect of the disclosed embodiments includes a real-time bidding system. The system includes a processor, and a memory. The memory includes instructions that, when executed by the processor, cause the processor to: receive user input data; generate a first machine learning model that generates a predicted expected performance based on the user input data; and adjust at least one bid on at least one of at least one keyword and at least one product associated with at least one marketplace, based on the predicted expected performance.

[0007] Another aspect of the disclosed embodiments includes a real-time bidding method. The method includes receiving user input data, generating a first machine learning model that generates a predicted expected performance based on the user input data, and adjusting at least one bid on at least one of at least one keyword and at least one product associated with at least one marketplace, based on the predicted expected performance.

[0008] These and other aspects of the present disclosure are disclosed in the following detailed description of the embodiments, the appended claims, and the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

[0010] FIG. 1 generally illustrates a schematic block diagram illustrating a system according to the principles of the present disclosure.

[0011] FIG. 2A is a schematic block diagram illustrating a computing device in the form of the smartphone of FIG. 1, according to the principles of the present disclosure.

[0012] FIG. 2B is a schematic block diagram illustrating a computing device in the form of the desktop computer of FIG. 1, and a server in the form of the first server of FIG. 1, according to the principles of the present disclosure.

[0013] FIG. 3 is a schematic block diagram illustrating a computing device and a server in operating a digital marketplace, according to the principles of the present disclosure.

[0014] FIG. 4 is a schematic block diagram illustrating a computing device and a server in hosting a digital marketplace that includes attributes of a target product and a competing product, according to the principles of the present disclosure.

[0015] FIG. 5 is a schematic block diagram illustrating a computing device that includes a graphic user interface according to the principles of the present disclosure.

[0016] FIG. 6 is a flow diagram illustrating a method of evaluating a product, according to the principles of the present disclosure.

[0017] FIG. 7 is a flow diagram illustrating a method of providing a competitive assessment of a target product on a marketplace, according to the principles of the present disclosure.

[0018] FIG. 8 is a schematic block diagram illustrating a computing device and a server in operating a digital marketplace, according to the principles of the present disclosure.

[0019] FIG. 9 is a graphic representation of a plurality of search terms plotted at points that represent a frequency and similarities in search terms associated with a target product relative to competing products, according to the principles of the present disclosure.

[0020] FIG. 10 is a graphic representation of a plurality of search terms plotted at points that represent relevance and volume of search terms associated with a target product relative to competing products, according to the principles of the present disclosure.

[0021] FIG. 11 is a graphical representation of a model output according to the principles of the present disclosure.

[0022] FIG. 12 is a flow diagram generally illustrating a real-time bidding method according to the principles of the present disclosure.

DETAILED DESCRIPTION

[0023] The following discussion is directed to various embodiments of the disclosure. Although one or more of these embodiments may be preferred, the embodiments disclosed should not be interpreted, or otherwise used, as limiting the scope of the disclosure, including the claims. In addition, one skilled in the art will understand that the following description has broad application, and the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to imply that the scope of the disclosure, including the claims, is limited to that embodiment.

[0024] In commerce systems, pay per click advertising operates on a cost-per-click auction, where bids are set at the target (e.g., keyword or product) level. Using the auction, bids are set to be competitive, and advertisements (ads) serve until the campaign budget cap is reached. Once the campaign budget cap is reached, advertising turns off and is no longer able to serve.

[0025] Typically, the industry standard protocol is to increase bids to spend more and decrease bids to spend less, often using a bidding algorithm that adjusts based on efficiency targets. Such bid adjustments happen without regard to the campaign budget and require manual inputs to spend a desired amount.

[0026] Accordingly, systems and methods, such as those described herein, configured improve real-time bidding, may be desirable. In some embodiments, the systems and methods described herein may be configured to provide optimal cost per click bidding which simultaneously or substantially simultaneously takes into account desired spend and target efficiency. The systems and methods

described herein may be configured to use machine learning, data extraction best practices, and data monitoring to create the optimal cost per click bidding system.

[0027] In some embodiments, the systems and methods described herein may be configured to maximize a user performance or success criteria, above a minimum threshold, while spending a user defined budget.

[0028] The systems and methods described herein may be configured to receive user input from a user computing device. For example, a user may interact with the user computing device to provide, at a user interface, input that includes the desired spend and the period for the desired spend amount. For example, the user can specify \$100 for a single day, week, month, etc. The user input may further include products and/or product information associated with the advertising budget.

[0029] The systems and methods described herein may be configured to receive, via a user selection made using the user interface, input indicating the marketplace(s) the budget, period, and products apply to. The systems and methods described herein may be configured to receive user selection indicating a minimum performance framework to fulfill per currency unit spent. For example, minimum \$5 in revenue per \$1 spent, minimum 2 new customers per \$1 spent, 300 user impressions per \$1 spent, etc. It should be understood that the user input may include any suitable input to fine tune or further direct the systems and methods described herein, such as specifying which groupings of target should receive different minimum performance requirements or budget expectations, or other suitable information.

[0030] The systems and methods described herein may be configured to, based on the user input for the specified time period, products, and marketplaces, estimate a distribution of relative expected advertising spend in order to determine whether spend is above, below, or on target. For example, if the time period (e.g., the user's specified time period) is a single day, 3 PM will generally have a higher anticipated spend than 3 AM, and the systems and methods described herein may be configured to recede from a machine learning model the relative spend throughout a day. The output of the model is a predicted distribution of user demand for each sub-period (e.g., hour) over the specified duration and marketplace.

[0031] In order to predict relative spend of each sub-period, the machine learning model uses a tree-based, neural network, or Bayesian architecture to ingest data, such as historical advertising spend by subdivided period, date and hour related elements, product and brand categories, demand behavior of similar products, advertising behavior of similar products, and/or the like. For example, if a user desires to spend \$100 for a single day on a marketplace for a group of products, given the product's history, category, brand, and other relevant information, the model anticipates that 5.3% of advertising spend would take place for the one-hour sub-period beginning at 9:00 AM. As live data arrives, the model continues to update and adjust to adapting conditions.

[0032] The systems and methods described herein may be configured to generate a population of potential targets to consider for each product. The systems and methods described herein may be configured to generate a significant quantity of potential targets, with their associated data and relevance, that meet a minimum filtering relevance for consideration in the process.

[0033] The systems and methods described herein may be configured to generate two models. The systems and methods described herein may be configured to use a first model of the two generated models, to estimate the performance metric given different potential bid levels. The systems and methods described herein may be configured to use a second model of the two generated models to estimate the relevant cost for each target given different potential bid levels. Additionally, or alternatively, the two models can be combined into a single model to generate output indicating performance per currency unit.

[0034] Each performance metric may include a ratio of performance per currency unit spent. The systems and methods described herein may be configured to maximize the performance ratio while meeting the desired spend, responsive to the actual spend being above the minimum threshold

ratio. For example, the systems and methods described herein may be configured to maximize revenue per dollar spent, while spending \$100 in a week.

[0035] The first model (e.g., the performance model) may output the expected performance, (e.g., \$50 of revenue), for any given bid and target. The first model may use a tree-based, neural network, Bayesian, or other architecture. The inputs to the first model may include: output metrics, the historical performance and bids of the target, historical performance of similar targets, product information, competitor information, category information, other target attributes, and/or the like.

[0036] The systems and methods described herein may be configured to fine tune the first model using the training inputs to meet the custom loss function according to user preferences captured through the interface and historical behavior. Specifically, advertising system risk preferences towards performance outcomes can be both asymmetric and change with magnitude. For example, a user may not be indifferent between an ROI of 0.1 over versus under a target, and a result that is 0.5 off a target may not simply be 5 times worse than a result that is 0.1. Accordingly, the first model is trained and fine tuned based on user preferences on performance. User preferences may be captured explicitly per user in direct questions or can be inferred from the preferences of the general population of users, such as average behavior.

[0037] The second model (e.g., the cost model) may output the expected cost in the specified currency unit, (e.g., \$10 of cost), for any given bid and target. Unlike the performance model, this is always cost rather than the user input specified. The second model may use a tree-based, neural network, Bayesian, or other architecture. The inputs to the second model may include: output metrics, the historical cost and bids for the target, historical performance of similar targets, product information, competitor information, category information, other target attributes, and/or the like.

[0038] The systems and methods described herein may be configured to fine tune the second model using training inputs to meet the custom loss function according to user preferences captured through the interface and historical behavior. Specifically, advertising system risk preferences towards spending outcomes can be both asymmetric and change with magnitude. For example, a user may not be indifferent between spending \$100 over target versus \$100 under a target, and a result that is \$500 off a target may not simply be 5 times worse than a result that is off by \$100. Accordingly, the second model may be trained and fine tuned based on user preferences on spending in both magnitude and direction.

[0039] The systems and methods described herein may be configured to combine the two models to generate both ratios and anticipated spend. For example, a \$0.50 bid for product ABC on target keyword “example” is estimated to produce \$3.00 in revenue and \$1.00 in cost, producing a revenue to cost ratio of 3. If the user requires a performance ratio of at least 4, then only lower bids, with associated lower revenue, lower cost, but higher ratios would be considered. As described, a single model to directly estimate the ratio may also be used depending on the conditions and accuracy. The final output of the model is a downward sloping curve (e.g., as is generally illustrated in FIG. 11) where, in aggregate across all targets, the total spend is the input and the ratio of performance per currency unit acts as the output, as depicted below. The systems and methods described herein may be configured to generate the output for any combination of period or sub-period, marketplaces, products, groups of products, etc. desired. Performance outcomes below the required performance threshold may be discarded.

[0040] In some embodiments, the systems and methods described herein may be configured to, knowing the desired amount of spend per sub-period, and an ability to select the collection of bids and targets that will achieve that desired spend at the highest performance, select all of the targets and bids associated with the desired spend and performance where the estimated spend that will sum to the desired spend. This collection of targets and bids is then sent into the marketplace for each sub-period.

[0041] The systems and methods described herein may be configured to provide real-time spend and performance monitoring throughout the period to achieve the desired user spend both above the

minimum performance and maximizing performance. The systems and methods described herein may be configured to incorporate the latest performance and spend into the models to generate updated estimates of performance and cost. As real-time performance of both spend and performance are generated, the latest data can be fed back into the models to learn through reinforcement and adjust to the latest market conditions and improve future predictions. The updated models can then make predictions for the remaining sub-periods of the day.

[0042] The systems and methods described herein may be configured to, based on the estimated model pacing on targeted spend, shift up or down the collecting of bids and targets throughout the period to achieve the closest possible spend to match the desired spend. For example, with a user target spend of \$100 for a 24 hour period, minimum performance metric is 4, 12 hours of the period has passed by, \$55 has been spent, the performance metric is 6, and the expected spend from the step two model is 44% (or \$44). In this scenario the systems and methods described herein may be spending over the expected target by \$11. First, the latest performance and spend may be incorporated into the models. Second, because spend is running over estimated target pacing, a selection of lower bids that corresponds with an estimated \$45 spend may be selected to complete the day in order to achieve the desired spend of \$55 (thus far)+\$45 (new estimate given updated targets, bids, and estimates)=\$100. This may result in higher estimated performance, as bids have become more conservative and performance per currency unit improves.

[0043] Referring to FIG. 1, a schematic block diagram illustrates a system **100** according to the principles of the disclosure. The system **100** may be used for the benefit of one or more users **110**, which may include a first user **112**, a second user **114**, a third user **116**, and a fourth user **118** as shown in FIG. 1. Each of the users **110** may use one of a variety of computing devices **120**, which may include any of a wide variety of devices that carry out computational steps, including but not limited to a desktop computer **122** used by the first user **112**, a laptop computer **124** used by the second user **114**, a smartphone **126** used by the third user **116**, a camera **128** used by the fourth user **118**, and the like. The system and method presented herein may be carried out on any type of computing device.

[0044] The computing devices **120** may optionally be connected to each other and/or other resources. Such connections may be wired or wireless, and may be implemented through the use of any known wired or wireless communication standard, including but not limited to Ethernet, 802.11a, 802.11b, 802.11g, and 802.11n, universal serial bus (USB), Bluetooth, cellular, near-field communications (NFC), Bluetooth Smart, ZigBee, and the like. In FIG. 1, by way of example, wired communications are shown with solid lines and wireless communications are shown with dashed lines.

[0045] Communications between the various elements of FIG. 1 may be routed and/or otherwise facilitated through the use of routers **130**. The routers **130** may be of any type known in the art, and may be designed for wired and/or wireless communications through any known communications standard including but not limited to those listed herein. The routers **130** may include, for example, a first router **132** that facilitates communications to and/or from the desktop computer **122**, a second router **134** that facilitates communications to and/or from the laptop computer **124**, a third router **136** that facilitates communications to and/or from the smartphone **126**, and a fourth router **138** that facilitates communications to and/or from the camera **128**.

[0046] The routers **130** may facilitate communications between the computing devices **120** and one or more networks **140**, which may include any type of networks including but not limited to local area networks such as a local area network **142**, and wide area networks such as a wide area network **144**. In some embodiments, the local area network **142** may be a network that services an entity such as a business, non-profit entity, government organization, or the like. The wide area network **144** may provide communications for multiple entities and/or individuals, and in some embodiments, may be the Internet. The local area network **142** may communicate with the wide area network **144**. If desired, one or more routers or other devices may be used to facilitate such

communication.

[0047] The networks **140** may store information on servers **150** or other information storage devices. As shown, a first server **152** may be connected to the local area network **142**, and may thus communicate with devices connected to the local area network **142** such as the desktop computer **122** and the laptop computer **124**. A second server **154** may be connected to the wide area network **144**, and may thus communicate with devices connected to the wide area network **144**, such as the smartphone **126** and the camera **128**. If desired, the second server **154** may be a web server that provides web pages, web-connected services, executable code designed to operate over the Internet, and/or other functionality that facilitates the provision of information and/or services over the wide area network **144**.

[0048] Referring to FIG. 2A, a schematic block diagram illustrates an exemplary computing device of the computing devices **120** that may enable implementation of the disclosure in a standalone computing environment. The computing device may be, for example, the smartphone **126** of FIG. 1. The present specification, however, contemplates that the computing device **120** may include any of those computing devices **120** described in FIG. 1 or any other type of computing device.

[0049] As shown, the smartphone **126** may include a processor **210** that is designed to execute instructions on data. The processor **210** may be of any of a wide variety of types, including microprocessors with x86-based architecture or other architecture known in the art, application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGA's), and the like. The processor **210** may optionally include multiple processing elements, or "cores." The processor **210** may include a cache that provides temporary storage of data incident to the operation of the processor **210**.

[0050] The smartphone **126** may further include memory **220**, which may be volatile memory such as random-access memory (RAM). The memory **220** may include one or more memory modules. The memory **220** may include executable instructions, data referenced by such executable instructions, and/or any other data that may beneficially be made readily accessible to the processor **210**.

[0051] The smartphone **126** may further include a data store **230**, which may be non-volatile memory such as a hard drive, flash memory, and/or the like. The data store **230** may include one or more data storage elements. The data store **230** may store executable code such as an operating system and/or various programs to be run on the smartphone **126**. The data store **230** may further store data to be used by such programs. For the system and method of the present disclosure, the data store **230** may store computer executable code associated with an assessment module **232**, a text analytics module **238**, a filtering module **235**, a comparison module **234**, a recommendation module **236**, and a competitiveness score generating module **233**. The data store **230** may further include data associated with descriptive terms **241** related to a target product and/or a competing product, relevant descriptive terms **242** associated with either of the target product or a competing product, a competitiveness score **239**, and an actionable report **237**. This data stored by the data store **230** may be maintained on the data store **230** for any length of time and some data may be created or overwritten at any time to facilitate the methods described herein.

[0052] The smartphone **126** may further include one or more wired transmitter/receivers **240**, which may facilitate wired communications between the smartphone **126** and any other device, such as the other computing devices **120**, the servers **150**, and/or the routers **130** of FIG. 1. The wired transmitter/receivers **240** may communicate via any known wired protocol, including but not limited to any of the wired protocols described in FIG. 1. In some embodiments, the wired transmitter/receivers **240** may include Ethernet adapters, universal serial bus (USB) adapters, and/or the like.

[0053] The smartphone **126** may further include one or more wireless transmitter/receivers **250**, which may facilitate wireless communications between the smartphone **126** and any other device, such as the other computing devices **120**, the servers **150**, and/or the routers **130** of FIG. 1. The

wireless transmitter/receivers **250** may communicate via any known wireless protocol, including but not limited to any of the wireless protocols described in FIG. **1**. In some embodiments, the wireless transmitter/receivers **250** may include Wi-Fi adapters, Bluetooth adapters, cellular adapters, and/or the like. Either of the wired transmitter/receiver(s) **240** or wireless transmitter/receiver(s) **250** may be associated with a network interface device (NID) **280**. The network interface device **280** may provide connectivity to, via the Internet, any network, e.g., a wide area network (WAN), a local area network (LAN), wireless local area network (WLAN), a wireless personal area network (WPAN), a wireless wide area network (WWAN), or other networks.

[0054] The smartphone **126** may further include one or more user inputs **260** that receive input from a user such as any of the users **110** of FIG. **1**. The users **110** described herein, may be referred to as a seller of a target product. The user inputs **260** may be integrated into the smartphone **126**, or may be separate from the smartphone **126** and connected to it by a wired or wireless connection, which may operate via the wired transmitter/receivers **240** and/or the wireless transmitter/receivers **250**. The user inputs **260** may include elements such as a touch screen, buttons, keyboard, mouse, trackball, track pad, stylus, digitizer, digital camera, microphone, and/or other user input devices known in the art.

[0055] The smartphone **126** may further include one or more user outputs **270** that provide output to a user such as any of the users **110** of FIG. **1**. The user outputs **270** may be integrated into the smartphone **126**, or may be separate from the smartphone **126** and connected to it by a wired or wireless connection, which may operate via the wired transmitter/receivers **240** and/or the wireless transmitter/receivers **250**. The user outputs **270** may include elements such as a display screen, speaker, vibration device, LED or other lights, and/or other output devices known in the art. In some embodiments, one or more of the user inputs **260** may be combined with one or more of the user outputs **270**, as may be the case with a touch screen. In some embodiments, the user outputs **270** may present to a user a graphical user interface by which the user may interact with the smartphone **126** in order to affect the methods and processes described herein.

[0056] The smartphone **126** may include various other components not shown or described herein. Those of skill in the art will recognize, with the aid of the present disclosure, that any such components may be used to carry out the present disclosure, in addition to or in the alternative to the components shown and described in connection with FIG. **2A**.

[0057] The smartphone **126** may be capable of carrying out the present disclosure in a standalone computing environment, i.e., without relying on communication with other devices such as the other computing devices **120** or the servers **150**. The present specification further contemplates that any of the assessment module **232**, competitiveness score generating module **233**, comparison module **234**, filtering module **235**, recommendation module **236**, and text analytics module **238** may be distributed amongst a number of computing devices (e.g., computing devices **120** of FIG. **1**) and/or amongst any server (e.g., **150** of FIG. **1**). In other embodiments, the present disclosure may be utilized in different computing environments. One example of a client/server environment will be shown and described in connection with FIG. **2B**.

[0058] Referring to FIG. **2B**, a schematic block diagram illustrates a computing device in the form of the desktop computer **122** of FIG. **1**, and a server in the form of the first server **152** of FIG. **1**, which may cooperate to enable practice of the disclosure with client/server architecture. As shown, the desktop computer **122** may be a “dumb terminal,” made to function in conjunction with the first server **152**.

[0059] Thus, the desktop computer **122** may have only the hardware needed to interface with a user (such as the first user **112** of FIG. **1**) and communicate with the first server **152**. Thus, the desktop computer **122** may include one or more user inputs **260**, one or more user outputs **270**, one or more wired transmitter/receivers **240**, and/or one or more wireless transmitter/receivers **250**. Again, either of the wired transmitter/receiver(s) **240** or wireless transmitter/receiver(s) **250** may be

associated with a NID **280a**. The NID **280a** may provide connectivity to, via the Internet, any network, e.g., a wide area network (WAN), a local area network (LAN), wireless local area network (WLAN), a wireless personal area network (WPAN), a wireless wide area network (WWAN), or other networks in which the first server **152** forms a part of. These components may be as described in connection with FIG. 2A.

[0060] Computing functions (apart from those incidents to receiving input from the user and delivering output to the user) may be carried out wholly or partially at the first server **152**. Thus, the processor **210**, memory **220**, data store **230**, wired transmitter/receivers **240**, and wireless transmitter/receivers **250** may be housed in the first server **152**. These components may also be as described in connection with FIG. 1A.

[0061] In operation, the desktop computer **122** may receive input from the user via the user inputs **260**. The user input may be delivered to the first server **152** via the wired transmitter/receivers **240** and/or wireless transmitter/receivers **250**. This user input may be further conveyed by any intervening devices, such as the first router **132** and any other devices in the local area network **142** that are needed to convey the user input from the first router **132** to the first server **152**.

[0062] The first server **152** may conduct any processing steps needed in response to receipt of the user input. Then, the first server **152** may transmit user output to the user via the wired transmitter/receivers **240**, and/or wireless transmitter/receivers **250**. This user output may be further conveyed by any intervening devices, such as the first router **132** and any other devices in the local area network **142** (or, alternatively, a wide area network **144**) that are needed to convey the user output from the first server **152** to the first router **132**. The user output may then be provided to the user via the user outputs **270**. In some embodiments, the user outputs **270** may present to a user a graphical user interface that, according to the methods described herein, display a listing of relevant descriptive terms **242** of the target product and competitive product as well as display an actionable report that describes a projected performance of the target product in a computer-networked marketplace relative to the at least one organic competing product also presented on the computer-networked marketplace.

[0063] Referring to FIG. 3, a schematic block diagram illustrating a computing device **322** (similar to any one of the computing devices shown in FIG. 1) and a server **350** (similar to any of the servers shown in FIG. 1) operating a digital marketplace, which may cooperate to enable practice of the disclosure with client/server architecture, according to the principles of the disclosure. As shown, the computing device **322** may be operatively coupled to the server **350** via the NID **380** as described herein. This operative coupling allows the computing device **322** to access, when appropriate, a digital marketplace **382** on which a target product and competitive product are sold. The digital marketplace **382** may be any network accessible website that lists a number of products that, when accessed by a user, allows a user to review products, rate products, purchase products among other tasks associated with digital commerce. The digital marketplace **382** may be managed by companies that include Ebay®, Amazon®, Wayfair®, Costco®, Walmart®, and Target®, among others. Upon purchase of a product, a consumer may have the purchased product sent to the consumer's home or business for consumption. In some embodiments, the digital marketplace **382** may be any of a plurality of websites that the server **350** provides storage and processing resources for.

[0064] As described herein, the computing device **322** may include a processor **310**, a memory **320**, user inputs **360**, user outputs **370** and a data store **330** that operate similar to those similar elements described in connection with FIGS. 2A and 2B. The data store **330** may include those modules described herein including an assessment module **332**, a competitiveness score generating module **333**, a comparison module **334**, a filtering module **335**, a recommendation module **336**, and a text analytics module **338**.

[0065] During operation, the assessment module **332** may assess certain attributes of a target product. The target product as described herein is a specific target product a user (e.g., seller) of the

computing device **322** is seeking to discover the competitiveness of the product within a certain market. For example, the target product may be a product the user is selling or would like to sell on the digital marketplace **382** hosted by the server **350**. In order to know the competitiveness of the target product, the assessment module **332** may access certain data about the target product present on the server **350**. The data may be accessed by the assessment module **332** by sending data requests via the NID **380** either via a wired (e.g., via the wired transmitter/receiver(s) **340**)) or a wireless (e.g., via the wireless transmitter/receiver(s) **350**) connection.

[0066] The data request may be a request for attributes regarding the target product. Although any number of attributes about the target product may be requested, the assessment module **332** may request specific attributes that will be used to develop an actionable report **337** regarding the competitiveness of the target in the digital marketplace **382**. A first attribute may be descriptive of the ratings provided by at least one purchaser of the target product on the digital marketplace **382**. Often, digital marketplaces **382** provide graphical user interfaces (GUIs) to consumers that allows those consumers to rate the products they purchase on the digital marketplace **382**. In a specific embodiment, a 5-star rating system may be used by a consumer/purchaser of the target product to rate the target product. A one-star rating would indicate a poor assessment by the consumer/purchaser of the target product while a 5-star rating would indicate a very good assessment of the target product by the consumer/purchaser. The assessment module **332** may, accordingly, take each star-rating or an average of those star-ratings as input for use in creating the actionable report **337**.

[0067] A second attribute may include the reviews and/or user generated content associated with the target product. Again, digital marketplaces **382** often provide a GUI that allow the consumer of the target product to enter text descriptive of the consumers' experiences with the target product. This text may include specific positive keywords or negative keywords that describe the consumers' experience with the target product. With this data, the assessment module **332** may cause a text analytics module **338** to, in some embodiments, parse each review for these keywords that describe the target product. Still further, the text analytics module **338** may also extract keywords descriptive of certain features of the target product. As an example, the wording "ergonomic handle" may be extracted by the text analytics module **338** describing not only that the target product includes a handle, but that that handle is an "ergonomic" handle giving a perception that the consumer giving that review likes the fit or feel of the target product.

[0068] A third attribute may be similar to the second attribute in that the assessment module **332** determines the number of the reviews associated with the target product presented on the digital marketplace **382**. The number of reviews may indicate a level of involvement with the target product either for the disparaging of the target product or the approval of the target product. Along with the textual substance of these reviews, the number of reviews associated with the target product may be used to help create the actionable report based on the involvement within the digital marketplace **382** with the target product.

[0069] A fourth attribute may include the listed price of the target product. Although the amount charged to purchase a product may not be indicative of the value of the target product, the charged amount relative to other similar competing products may be indicative of its worth or current price point (whether incorrect or correct).

[0070] A fifth attribute may also include a ranking of the target product relative to at least one organic competing product. This ranking may be a result of an average or accumulative rating of the target product relative to the organic competing product. Often, the digital marketplaces **382** allow purchasers to list organic competing products and the target product by an average rating. By doing so the assessment module **332** may understand the ranking of the target product relative to the at least one organic competing product and use this information to develop the actionable report **337**.

[0071] The assessment module **332** may also determine similar attributes of an at least one organic

competing product similar to those attributes discovered by the assessment module **332** for the target product. In the context of the present specification the term “organic competing product” is meant to be understood as any product that, based on consumer reviews, is ranked on the digital marketplace **382**. An “organic” competing product is therefore a naturally ranked product based on those reviews provided by past consumers as opposed to those products that may be given “top shelf” preference after payment to achieve such status. This organic ranking nature of products on the digital marketplace **382** is often done to provide potential consumers with evidence that others appreciate that product. A “competing” product is any product that is similar to the target product but sold by another seller apart from the seller of the target product. The “similarity” of the target product relative to the at least one organic competing product is dependent on the data obtained by the text analytics module **338** and specifically the analysis of descriptive terms **341** associated with each of these types of products. In a specific embodiment, the text analytics module **338** may also obtain descriptive data associated with each target product and organic competing product per their listing. Again, digital marketplaces **382** allow descriptions of products to be posted alongside each product that describes its functionalities, its physical characteristics, and its alleged advantages as superior products. All of this is presented to a potential consumer on a GUI as textual information used to entice the consumer to purchase the products. The text analytics module **338** may analyze this text and, using a parsing process, extract keywords used to compare the text associated with the target product to the text associated with the organic competing product.

[0072] When the computing device **322**, via the assessment module **332**, has obtained the attributes associated with the target product and the at least one organic competing product, the descriptive terms **341** describing these attributes may be listed for consumption by, in some embodiments, a filtering module **335**. The filtering module **335** may be used to filter the descriptive terms **341** to only those relevant descriptive terms **342** that have resulted in the purchase of the target product in the digital marketplace **382**. For example, some descriptive terms **341** may, rightly or wrongly, include a color or color scheme of the target product or organic competing product. Although some consumers may appreciate a specific color of a product, these may not be deciding factors used to entice a consumer to purchase the target product or organic competing product. This may be especially true where, as indicated by purchase histories associated with the target product or organic competing product indicate that any particular color of product was not overwhelmingly purchased over another color. In this specific example, although the color of the product is a descriptive term **341** the text analytics module **338** had parsed out from the products, it may not necessarily be a relevant descriptive term **341** and such information may be filtered out by the filtering module **335** to obtain only those relevant descriptive terms **342** associated with any of the target product or organic competing product.

[0073] In a more general example, the filtering module **335** may narrow down the descriptive terms **341** of interest by analyzing metrics collected on sufficiently “mature” keywords (e.g., sales >2) as budding keywords that may lack sufficient data to influence predictions in purchasing the target product or organic competing product. The click-rate and conversion rate (clicks that result in a purchase) associated with any given product may be taken into consideration based on the keywords used to search for the products. In these examples, a lack of data regarding a specific descriptive term **341** may also filter out that specific descriptive term **341** in order to obtain the relevant descriptive terms **342** as described herein. It is also appreciated that the descriptive terms **341** may be filtered by the filtering module **335** based on any other reason to obtain relevant descriptive terms **342** and the present specification contemplates these other reasons.

[0074] With the relevant descriptive terms **342** being determined, these relevant descriptive terms **342** may be sent to a comparison module **334** to compare those relevant descriptive terms **342** of the target product to those relevant descriptive terms **342** associated with the at least one organic competing product. Although the present specification describes this comparison process as being conducted between a single organic competing product (e.g., “at least one”) to the target product,

any number of organic competing products may be compared to the target product. In a specific example, the top 10 ranked organic competing products may be compared to the target product by the comparison module **334**.

[0075] During execution of the comparison module **334** by the processor **310**, the descriptive terms **341** may be compared to generate, with a competitiveness score generating module **333** executed by the processor **310**, a competitiveness score **339**. In some embodiments, the competitiveness score may use any process or algorithm used to define how the target product can or cannot compete with any of the discovered organic competing products.

[0076] During operation, a recommendation module **336** may receive this competitiveness score **339** along with other data from the digital marketplace **382** hosted by the server **350**. Among this other data may include revenue data associated with the organic competing products and the target product (if available). For example, where a click-rate of any given product (e.g., target product or organic competing product) results in a purchase, this conversion rate data along with the pricing data of the products may be passed to the recommendation module **336**. The recommendation module **336** may then provide a recommendation descriptive of the ability (or inability) of the target product to compete with the at least one organic competing product. In some embodiments, a threshold competitiveness score may be set such that the report provided by the recommendation module **336** indicates to the seller of the target product whether to proceed to sell that product on the digital marketplace **382**. Alternatively, where the competitiveness score has not met the threshold the competitiveness score generating module **333** may not forward the competitiveness score onto a recommendation module **336** to generate the actionable report **337**.

[0077] Alternatively, or additionally, where the competitiveness score has not met the threshold the competitiveness score generating module **333** may pass a threshold failure signal onto to the recommendation module **336** indicative of a non-competitive status of the target product. When the threshold competitiveness score is not reached, the recommendation module **336** may provide an indication to the seller that it is not recommended that the seller initiate or continue to sell the target product on the digital marketplace **382**.

[0078] Where the threshold competitiveness score is reached, the recommendation module **336** may provide additional economic data descriptive of price points and ACoS statistics to use in order to increase revenue. Again, a seller of the target product may not know what appropriate target advertising cost of sale (ACoS) to meet and what price point to sell the target product at in order to see long term gains in lieu of short-term profits. The recommendation module **336** provides this information based on the competitiveness score **339** generated by the competitiveness score generating module **333** and revenue data received from the digital marketplace **382**. In a specific example, the revenue potential of the target product may be determined by the recommendation module **336** calculating an ad spend margin, an ad spend potential, and a revenue potential. The ad spend margin may be calculated by multiplying a target ACoS by the price of the target product. A target ACoS may be determined and set by the seller based on available capital or may be set by the seller based on the fraction of the revenue received thus far from the sale of the target product on the digital marketplace **382** and costs of manufacturing. Ad spend potential may then be calculated by multiplying monthly opportunity units (OU) by the spend margin. The monthly OUs may be calculated as a result of the conversion rate of clicks to the target product that is the result of sales of the target product after a purchaser has viewed the product. The revenue potential may then be calculated by multiplying the OU with the price of the target product. This revenue potential of each of the target products and organic competing products may be ranked to determine the placement of the target product within the digital marketplace **382**.

[0079] In some embodiments, the recommendation (e.g., the actionable report **337**) presented by the recommendation module **336** may be refined by inputting an estimated bid amount from the digital marketplace **382** required to “win” advertising slots for the target product. The digital marketplace **382**, along with selling products, may also engage in presenting advertisements to a

potential purchaser of one or more products. These advertisements may be presented in a banner or other sub-section of the GUI presented to the purchaser or as a pop-up window advertisement. These forms of advertisements present, in real-time, alternative products for which the potential purchaser is seeking to purchase. These advertisements may present the target product and persuade the purchaser to purchase the target product rather than a competitors' products. Thus, investments may be required to increase the purchasing instances of the target product. The present systems and methods may also present to the seller of the target product, on the actionable report **337**, how much additional investment may be needed to win advertising slots based on the keywords associated with the target product and entered into a search by a potential user. For example, the investment needed may be calculated by multiplying the projected bid amount by the product of the click rate of the target product and the impressions (e.g., uses) for specific keywords associated with the target product and the organic competing product used to search for those products. A return on investment (ROI) may then be calculated by subtracting the investment needed from an investment payoff term and multiplying that by the ad spend potential. Products with no (or low) potential receive suggestion outputs as to why they are not competitive or have bad conversion rates by the recommendation module **336** and its actionable report **337**, so that these attributes of the target product can be improved for future potential or the money spent to sell the target product can be reallocated for other uses.

[0080] FIG. **4** is a schematic block diagram illustrating a computing device **420** and a server **452** in hosting a digital marketplace **482** that includes attributes of a target product and a competing product, which may cooperate to enable practice of the disclosure with client/server architecture. As described herein, the assessment module **432** may assess certain attributes of a target product. The target product as described herein is a specific target a user (e.g., seller) of the computing device **420** is seeking to discover the competitiveness of the product within a certain market. For example, the target product may be a product the user is selling or would like to sell on the digital marketplace **482** hosted by the server **452**. In order to know the target product's competitiveness, the assessment module **432** may access certain data about the target product present on the server **452**. The data may be accessed by the assessment module **432** by sending data requests via the NID **480** either via a wired (e.g., via the wired transmitter/receiver(s) **440**) or a wireless (e.g., via the wireless transmitter/receiver(s) **450**) connection.

[0081] The data request may be a request for attributes regarding the target product. Although any number of attributes about the target product may be requested, the assessment module **432** may request specific attributes that will be used to develop an actionable report regarding the competitiveness of the target in the digital marketplace **482**. A first attribute may be descriptive of the ratings **483** provided by at least one purchaser of the target product on the digital marketplace **482**. Often, digital marketplaces **482** provide graphical user interfaces (GUIs) to consumers that allows those consumers to rate the products they purchase on the digital marketplace **482**. In a specific embodiment, a 5-star rating system may be used by a consumer/purchaser of the target product to rate the target product. A one-star rating would indicate a poor assessment by the consumer/purchaser of the target product while a 5-star rating would indicate a very good assessment of the target product by the consumer/purchaser. The assessment module **432** may, therefor, take each star-rating or an average of those star-ratings as input for use in creating the actionable report.

[0082] A second attribute may include the content **486** of the reviews and description associated with the target product. Again, digital marketplaces **482** often provide a GUI that allow the consumer of the target product to enter text descriptive of the consumers' experiences with the target product. This text may include specific positive keywords or negative keywords that describe the consumers' experience with the target product. With this data, the assessment module **432** may cause a text analytics module **438** to, in some embodiments, parse each review that describes the target product. Still further, the text analytics module **438** may also extract keywords descriptive of

certain features of the target product. As an example, the wording “ergonomic handle” may be extracted by the text analytics module **438** describing not only that the target product includes a handle, but that that handle is an “ergonomic” handle giving a perception that the consumer giving that review likes the fit of the target product.

[0083] A third attribute may be the number of the reviews **484** associated with the target product presented on the digital marketplace **482**. The number of reviews **482** may indicate a level of involvement with the target product either for the disparaging of the target product or the approval of the target product. Along with the textual substance of these reviews, the number of reviews associated with the target product may be used to help create the actionable report based on the involvement within the digital marketplace **482** with the target product.

[0084] A fourth attribute may include the listed price **485** of the target product. Although the amount charged to purchase a product may not be indicative of the value of the target product, the changed amount relative to other similar competing products may be indicative of its worth or current price point (whether incorrect or correct).

[0085] A fifth attribute may also include a ranking **487** of the target product relative to at least one organic competing product. This ranking may be a result of an average or accumulative rating of the target product relative to the organic competing product. Often, the digital marketplaces **382** allow purchasers to list organic competing products and the target product by an average rating. By doing so the assessment module **432** may understand the ranking of the target product relative to the at least one organic competing product and use this information to develop the actionable report.

[0086] Each of these target product attributes may be requested by the computing device **420** and its assessment module **432** and delivered by the server **452** upon request. Even further, similar attributes related to at least one organic competing product may also be requested by and sent to the computing device **420**. These organic product attributes may include competing product ratings **488**, competing product review numbers **489**, competing product prices **490**, competing product content **491**, and competing product rank **492**. Each of these competing product attributes may be similar to those attributes associated and described herein in connection with the target product.

[0087] FIG. 5 is a schematic block diagram illustrating a computing device **520** that includes a graphic user interface **522** used to enable practice of the disclosure within a client/server architecture. The graphic user interface **522** may be used by a seller of a target product to evaluate the competitiveness of the target product as described herein. As described herein, the computing device **520** includes a filtering module **535**. The filtering module **535** may be used to filter the descriptive terms **541** to only those relevant descriptive terms **542** that have resulted in the purchase of the target product in the digital marketplace.

[0088] The filtering module **535** may include a number of types of filters to filter the descriptive terms **541** into the relevant descriptive terms **542**. These filters may include an impression filter **524**, a click-rate filter **526**, and a conversion-rate filter **528** each of which may result in the removal of descriptive terms **541** that do not result in purchases of the target product or any organic comparison product. As described herein, the impression filter **524** may be provided with a number of times an ad associated with the target product or competing product (whether it's a banner, button, or text link) has been (or will be) exposed to a potential purchaser and has resulted in a purchase of that product. The impression filter **524** may therefore, filter out those instances where a potential purchaser did not see or was not shown an ad but did result in a purchase. Click-rate filter **526** may filter out those descriptive terms that, despite the wording of the ad, did not result in a selection of the ad or a purchase of the product. The conversion-rate filter **528** may filter out those descriptive terms that, despite the wording of the ad and a selection by the potential purchaser of the ad, did not result in a purchase of the product.

[0089] By filtering the descriptive terms via the filtering module **535** and its associated filters **524**, **526**, **528**, the GUI **522** may be able to display to a seller of the target product those relevant

descriptive terms **542** that apply in the analysis of how competitive the target product is. Although FIG. **5** shows the use of specific filters **524**, **526**, **528** to filter the descriptive terms **541**, the present specification contemplates that the descriptive terms **541** may be filtered using any criteria. [0090] In some embodiments, the system **100** and/or any computing device or other devices or systems described herein, may perform the methods described herein. However, the methods described herein as performed herein are not meant to be limiting, and any type of software executed on a controller or processor can perform the methods described herein without departing from the scope of this disclosure.

[0091] FIG. **6** is a flow diagram illustrating a method **600** of evaluating a product, according to the principles of the disclosure. The method **600** may begin at block **605** with assessing attributes of a target product using an assessment module executed by a processor. As described herein, the assessment of the target product (or any other competing product) may indicate certain attributes of the target product. Although any number of attributes about the target product may be requested, the assessment module may request specific attributes that will be used to develop an actionable report regarding the competitiveness of the target in the digital marketplace.

[0092] At block **610**, the method **600** may further include listing relevant descriptive terms of the target product descriptive of the attributes of the target product. This listing of the relevant descriptive terms may also be conducted by the assessment module being executed by the processor of the computing device. This list of relevant descriptive terms, in some embodiments, may have been generated based on the filtering of all descriptive terms generated for the target product as described herein. There may be some irrelevant information that may be filtered out of the descriptive terms generated from the attributes of the target product that would not need to show up in the actionable report.

[0093] The method **600** may continue at block **615** with accessing a computer-networked marketplace, via a NID, and identifying at least one organic competing product matching at least one descriptive term. This identification may implement the assessment module to compare the descriptive terms associated with the target product to any generated descriptive terms associated with any organic competing product. In some embodiments, this matching process of descriptive terms related to the target product to descriptive terms related to the organic competing product may be conducted before or after the filtering of descriptive terms by a filtering module as described herein. When conducted before, more organic competing products may be matched where, when conducted after the filtering, relatively less organic competing products may be matched due to the smaller list of relevant descriptive terms.

[0094] The method **600** may also include comparing the descriptive terms of the target product to descriptive terms associated with the at least one organic competing product to generate a competitiveness score at block **620**. This may be done via execution of a comparison module **620** executed by the processor. During execution of the comparison module by the processor, the descriptive terms may be compared to generate, with a competitiveness score generating module executed by the processor, a competitiveness score. In some embodiments, the competitiveness score may use any process or algorithm used to define how the target product can or cannot compete with any of the discovered organic competing products.

[0095] At block **625**, the method **600** may further include generating an actionable report descriptive of a projected performance of the target product in the computer-networked marketplace relative to the at least one organic competing product. The actionable report may be generated via the execution of a recommendation module by the processor. During operation, a recommendation module may receive this competitiveness score along with other data from the digital marketplace hosted by the server. Among this other data may include revenue data associated with the organic competing products and the target product (if available). For example, where a click-rate of any given product (e.g., target product or organic competing product) results in a purchase, this conversion rate data along with the pricing data of the products may be passed to the

recommendation module. The recommendation module may then provide a recommendation descriptive of the ability (or inability) of the target product to compete with the at least one organic competing product. In some embodiments, a threshold competitiveness score may be set such that the report provided by the recommendation module **336** indicates to the seller of the target product whether to proceed to sell that product on the digital marketplace. Alternatively, where the competitiveness score has not met the threshold the competitiveness score generating module may not forward the competitiveness score onto a recommendation module to generate the actionable report. When the threshold competitiveness score is not reached, the recommendation module simply provides an indication to the seller that it is not recommended that the seller initiate or continue to sell the target product on the digital marketplace.

[0096] Where the threshold competitiveness score is reached, the recommendation module may provide additional economic data descriptive of price points and ACoS statistics to use in order to increase revenue. Again, a seller of the target product may not know what appropriate target ACoS to meet and what price point to sell the target product at in order to see long term gains in lieu of short-term profits. The recommendation module provides this information based on the competitiveness score generated by the competitiveness score generating module and revenue data received from the digital marketplace. In a specific example, the revenue potential of the target product may be determined by the recommendation module calculating an ad spend margin, an ad spend potential, and a revenue potential. The ad spend margin may be calculated by multiplying a target ACoS by the price of the target product. A target ACoS may be determined and set by the seller based on available capital or may be set by the seller based on the fraction of the revenue received thus far from the sale of the target product on the digital marketplace and costs of manufacturing. Ad spend potential may then be calculated by multiplying monthly opportunity units (OU) by the spend margin. The monthly OUs may be calculated as a result of the conversion rate of clicks to the target product that is the result of sales of the target product after a purchaser has viewed the product. The revenue potential may then be calculated by multiplying the OU with the price of the target product. This revenue potential of each of the target products and organic competing products may be ranked to determine the placement of the target product within the digital marketplace.

[0097] At this point, the method **600** may end.

[0098] FIG. 7 is a flow diagram illustrating a method **700** of providing a competitive assessment of a target product on a marketplace, according to the principles of the disclosure. Here, the method **700** may begin with evaluating a target product to determine attributes of the target product at block **705**. In some embodiments, the evaluation may be conducted via the execution of an assessment module. In some embodiments, the assessment may be conducted by requesting, at a GUI, descriptive terms regarding the target product. Additionally, or alternatively, the evaluation may be made by an assessment module accessing a digital marketplace to retrieve descriptive terms via a text analytics module as described herein. Additionally, or alternatively, certain input devices such as a digital camera may be used to image the target product and extrapolate certain features of the product such as size, color, texture, among others.

[0099] The method **700** may continue at block **710** with accessing the digital marketplace to determine at least one organic competing product to the target product upon execution of the processor. In this embodiment, the assessment module may access certain data about the target product such as the descriptive terms and cross-reference those descriptive terms to determine if at least one descriptive term matches any competing product listed on the digital marketplace.

[0100] At block **715**, the method **700** may include calculating a competitiveness score related to the ability of the target product to compete with the at least one organic competing product. This process may be conducted upon execution of a competitiveness score generator by the processor of the computing device accessing the digital marketplace. In some embodiments, the competitiveness score may use any process or algorithm used to define how the target product can or cannot

compete with any of the discovered organic competing products.

[0101] The method **700** may further include generating an actionable report based on the ability of the target product to compete with the at least one organic competing product at block **720**. During operation, a recommendation module, executed by the processor, may receive the competitiveness score along with other data from the digital marketplace hosted by the server. Among this other data may include revenue data associated with the organic competing products and the target product (if available). For example, where a click-rate of any given product (e.g., target product or organic competing product) results in a purchase, this conversion rate data along with the pricing data of the products may be passed to the recommendation module. The recommendation module may then provide a recommendation descriptive of the ability (or inability) of the target product to compete with the at least one organic competing product. In some embodiments, a threshold competitiveness score may be set such that the report provided by the recommendation module indicates to the seller of the target product whether to proceed to sell that product on the digital marketplace. Alternatively, where the competitiveness score has not met the threshold the competitiveness score generating module may not forward the competitiveness score onto a recommendation module to generate the actionable report. When the threshold competitiveness score is not reached, the recommendation module simply provides an indication to the seller that it is not recommended that the seller initiate or continue to sell the target product on the digital marketplace. At this point, the method **700** may end.

[0102] FIG. **8** is a schematic block diagram illustrating computing device **822** and a server **852** in operating a digital marketplace **882**, which may cooperate to enable practice of the disclosure with client/server architecture. In addition to providing an actionable report (FIG. **3**, **337**) regarding the competitiveness of the target in the digital marketplace **882** as described in connection with FIGS. **1-7**, the present computing device **822** may further describe an actionable report **837** that describes sustainable and feasible growth over time on an ecommerce platform (e.g., the digital marketplace **882**) on a product level as well as provide a winnability report **804** descriptive of a probability of winning each search term (e.g., having the target product associated with the search term) at any given point in time along with the estimated costs to win those search terms. The actionable report **837** and winnability report **804** may, in some embodiments, provide a user with an indication as to how to optimize advertising and search engine implementation to increase revenue.

[0103] As described herein, the computing device **822** may include a processor **810**, a memory **820**, user inputs **860**, user outputs **870** and a data store **830** that operate similar to those similar elements described in connection with FIGS. **2A** and **2B**, for example. The data store **830** may include those modules described herein including a comparison module **834**, and a revenue module **899**.

[0104] The computing device **822** described may include any module, data store **830**, or data maintained on the computer as those described in connection with FIG. **3** herein. In the embodiments described herein, an actionable report **837** may be provided using a comparison module **834** similar to the comparison module **334** described in connection with FIG. **3**. Although these modules (e.g., comparison module **834**) may be similar to those described in FIG. **3**, the modules in FIG. **8** may perform additional and different processes as described herein in order to provide an actionable report **837** indicating optimized advertising and search engine implementation.

[0105] In some embodiments, the computing device **822** may initially determine any competitive products that, at any point in time, compete with the target product. The computing device **822** may do this by accessing a search engine **894** associated with a digital marketplace **882** via the processor **810** and NID **880** of the computing device **822**. Upon accessing the search engine **894**, the processor **810** may retrieve data descriptive of the frequency of appearance of one or more search terms associated with the target product. Additionally, the processor **810** may obtain data related to the ranking of those search terms. This data may be descriptive of the coincidence that the target product and any competitive product are associated with the same search terms. Still

further, this data may be descriptive of how the search terms associated with the target product and each competitive product are similar in their rankings. For example, where the target product is an athletic shoe, some pertinent search terms may include running, hiking, basketball, tennis, sole, laces, and marathon among other potential terms associated with the target product athletic shoe. The data may also include which competing products also rank similarly with these terms. For example, a competing product that matches 9 out of 10 search terms with the target product is “higher ranked” as compared to a competing product that matches 4 out of 10 search terms.

[0106] In a specific embodiment, the processor **810** may access this data using, for example, a search query website such as Google® Trends®. These types of websites may be used by the processor **810** to access a number of search queries for specific terms associated with any of the target product and any number of competitive products. The search query websites may be accessed by the processor **810** to automatically access search query inquiries in order to obtain the data used herein by the computing device **822**. Although specific search query websites are contemplated herein, the present specification also contemplates that other search query databases may be accessed by the processor **810** whether those databases are accessible by a user via a website or not.

[0107] The computing device **822** also includes a machine learning module **896** (e.g., which may be referred to herein as the machine learning model **896**). The machine learning module **896** may build a number of mathematical models that provide a competitive set report **898** describing a competitive set of products that compete with the target product. As with each machine learning module **896**, the machine learning module **896** may be “taught” by using, as input, a plurality of sets of target product search terms and rankings as well as a plurality of sets of competing product search terms and rankings. Again, the plurality of sets of target product search terms and rankings as well as a plurality of sets of competing product search terms and rankings may be accessible by the processor **810** either via a specific search query website or database.

[0108] The machine learning module **896**, in some embodiments, may, upon execution by the processor **810**, determine such correlations, in some embodiments, based on any machine learning or neural network methodology known in the art or developed in the future. In a specific embodiment, the machine learning module **896** may implement an unsupervised learning clustering technique. For example, the machine learning module, in some embodiments, may model the relationships between each plurality of sets of target product search terms and rankings as well as a plurality of sets of competing product search terms and rankings using a layered neural network topology. Such a neural network, in some embodiments, may include an input layer (e.g., plurality of sets of target product search terms and rankings as well as a plurality of sets of competing product search terms and rankings) including a known, recorded set of values for each of these parameters, settings, indicators, and usage data metrics, and an output layer including a projected optimal competitive set report **898**, based on the known, recorded set of values in the input layer. The machine learning module **896**, in some embodiments, may propagate input through the layers of the neural network to project or predict optimal competitive set report **898** based on the known and recorded search term metrics, and compare these projected values to optimal search terms to be presented in the competitive set report **898**. Using a back-propagation method, the machine learning module **896**, in some embodiments, may then use the difference between the projected values and the known optimal values to adjust weight matrices of the neural network describing the ways in which changes in each of the search term data metrics are likely to affect the optimal search terms to be presented in the competitive set report **898**.

[0109] With the output layer, the computing device **822** may provide learned competitive search terms that are determined to be the optimal search terms if any have been designated and based upon the similar and frequent search terms detected at the search engine **894** of the digital marketplace **882** during use of the computing device **822**. These resulting learned optimal search terms may be suggested to a user or automatically implemented. Suggestion may come with an

indicator and may be shown in a graph at a user interface (e.g., such as a graphical user interface) for, in some embodiments, approval by the user before implementation of the other processes executed by the processor **810** of the computing device **822**.

[0110] An example representation of the graph is shown in FIG. **9**. This example graph may indicate positions of each search term of a competitive product relative to the target product based on the frequency. Each point (e.g. circle) on the graph represented in FIG. **9** is representative of a search term. Each representative search term is arranged on the graph in FIG. **9** at a point that defines that terms frequency in appearing together with a search term of the target product and at a position where the search term is similar or not relative to the search terms associated with the target product. In this example graph, the further to the right any given search term is, the more similar the search terms of a competitive product are similar to the search terms of the target product. Additionally, the further to the left any given search term is, the less similar the search terms of the competitive product are similar to the search terms of the target product. Further, the closer to the top of the graph any given search term is, the more general the search term is compared to the target product while the closer to the bottom of the graph any given search term is, the more niche the search term is compared to the target product. In some embodiments, it may be most desirable to have a target product that has associated search terms relative to the search terms of a competitive product that is more general and similar. This indicates that the target product is competing with relatively well-known competing products. The processes described herein, may help to provide a report to a user indicative of how to adjust advertisement revenue to focus on more general and similar search terms as the competitive products.

[0111] In some embodiments, the machine learning module **896** may perform a forward propagation and backward propagation, using different input node values repeatedly to finely tune any matrices either weighted or not. In such a way, the machine learning module **896**, in some embodiments, may adaptively learn how changes in the plurality of sets of target product search terms and rankings as well as a plurality of sets of competing product search terms and rankings may affect the data reflected in the competitive set report **898**. The weight matrices associated with the layers of the neural network model in such an embodiment may describe, mathematically, these correlations for an individual target product. The neural network model (including designation of the node values in the input layer, and number of layers), along with the weight matrices associated with each layer, in some embodiments, may form a trained machine learning classifier, algorithm, or mathematical model to be used in generating any competitive set report **898** as described herein.

[0112] As described herein, the output from the, now trained, machine learning module **896** is a competitive set report **898**. With the competitive set report **898** the computing device **822** may, with the processor **810** and NID **880**, determine a current performance on the search terms related to the target product that are most relevant to the competitive set defined in the competitive set report **898**. In this process, the two variables that are discovered are how often a term appears in a search generally (e.g., a general search term volume, or how many times people search the term per day) and how often the term appears in searches associated with the competitive set report **898**. More specifically, in some embodiments, those search terms found to be most general and similar among the target product and each competitive product are provided to the comparison module **834** which searches, via execution of the processor **810** at the search engine **894**, those search terms defined in the competitive set report **898**. During this process, the processor **810** may access the search engine **894** at the digital marketplace **882** or any other search engine and obtain search term metadata that describes the current performance of each of the search terms related to the target product that are most relevant to the competitive set defined in the competitive set report **898**. The comparison module **834** may compare these most relevant search terms from the competitive set report **898** and provide that data to the user in the form of an actionable report **837**. In some embodiments, the data descriptive of the search terms related to the target product that are most relevant to the competitive set in the actionable report **837** may be provided to the user via a graphical representation.

[0113] An example graphical representation of this current performance on the search terms related to the target product is shown in FIG. 10. As shown in FIG. 10, the further to the right of the graph any search term (e.g., represented by a circle) is, the search term has a higher volume or appears more often than the other search terms indicating a relatively higher relevance to competing products. Additionally, the further to the left of the graph any search term is, the search term has a lower volume or appears less often than the other search terms indicating a relatively lower relevance to competing products. Also, the further to the top of the graph any search term is, the search term has a higher relevance than the other search terms indicating a relatively higher relevance to competing products. Further as the search term is placed lower on the graph, the search term has a lower relevance than the other search terms indicating a relatively lower relevance to competing products. The most frequently searched and relevant terms may be provided to the comparison module **834** as well and used to further define the sustainability and feasible growth over time of the target product on, for example, the digital marketplace **882**.

[0114] With those most relevant and most frequent search terms as indicated in FIG. 10 being discovered and presented in the actionable report **837**, the computing device **822** may also quantify an opportunity of those search terms that, when associated with the target product, would increase the revenue and profit margins in selling the target product. In some embodiments, the processor **810** may execute a revenue module **899** to receive those relevant and most frequent search terms from the actionable report **837** and provide output to a user in the form of an increased revenue metric. The increase revenue may be calculated by the revenue module by, upon execution of the processor **810**, the following formula:

Increased Revenue=Impressions*Click Rate*Conversion Rate*Basket Size*Price Equation 1

[0115] In the context of Equation 1, the impressions may be defined as the search volume of each of those most relevant and most frequent search terms, in some embodiments. In some embodiments, the quantity of impressions may be measured by a number of times an ad associated with the target product is presented to any given user during or after those most relevant and most frequent search terms are entered into a search engine **894**. This data may be retrieved by the processor **810** by accessing a particular database or, as described herein, accessing a search query website.

[0116] In some embodiments, the click rate of Equation 1 may be defined as an estimation along a curve of the probabilities of receiving clicks associated with the rank for each of the most relevant and most frequent search terms provided by the actionable report **837**. For example, a ranking may be set to include a first place click rate (e.g., 20% of clicks), second place click rate (14% of clicks), up until a 10th place click rate (6% of clicks) and beyond to any number of ranked most relevant and most frequent search terms. This data may be retrieved by the processor **810** by accessing a particular database or, as described herein, accessing a search query website.

[0117] The conversion rate in Equation 1 may, in some embodiments, be defined as percentage of those most relevant and most frequent search terms that were clicked and associated with the target product and converted into a sale (e.g., resulted in a sale of the target product). This data may be retrieved by the processor **810** by accessing a particular database or, as described herein, accessing a search query website.

[0118] In some embodiments, the basket size may be defined as the number of units purchased with each conversion. This number may be averaged over a plurality of purchases, in some embodiments. For example, where a number of conversions have been detected, the processor **810** may calculate how many units of the target product were purchased at any one time (e.g., units placed in a “shopping cart” for purchase at the digital marketplace **882**). This value may at least be equal to 1 or more. Again, this data may be retrieved by the processor **810** by accessing a particular database or, as described herein, accessing a search query website.

[0119] The price of the target product may be, in some embodiments, a suggested retail price by the

manufacturer. In some embodiments, the quantitative value of the price in Equation 1 is an average price of the target product, or other brand products resulting from the purchase, across any plurality of digital marketplaces **882** net of any discounts or promotions associated with those sales. This data may be retrieved by the processor **810** by accessing a particular database, accessing a search query website as described herein, or accessing sales data from a database maintained by the manufacturer of the target product.

[0120] In some embodiments, any of the impression values, click rate values, conversion rate values, basket size values, and price values in Equation 1 may be augmented by a weight value. In this embodiment, the weight value may accentuate or abate the effect of any one of these values in Equation 1 in order to better determine an increased revenue value or opportunity by the seller of the target product to increase that revenue. Because the actual, real-time data is being used in Equation 1, the seller of the target product or user of the computing device **822** may know, in real-time, whether to take advantage of any instance of increased views or sales of a product in order to increase interest in the target product over any competitors' products.

[0121] In some embodiments, the value associated with click rate in Equation 1 may significantly shift a decision by a user of the computing device **822** whether to take an action such as provide more advertising supporting the target product. This click rate associated with improving the search rank from the target product's current position on a search term to a potential rank position of a search phrase may be weighted to accommodate for an increase in importance of this value in some embodiments. For example, for a given search term that may improve an organic search rank for any of the search terms from 20th rank to 5th rank will improve the click rate by an estimated 3 times. Some of the improvement in rank may also originate from increased impressions and especially in situations where having an unranked target product on a search term achieves a search rank 10th among the rankings. In this example, this would improve clicks from zero (due to zero impressions) to the associated estimated clicks of 10th rank on that search term. As output, the processor **810** may, via the revenue module **899**, provide an increased revenue report **802** describing how to, if at all, increase the revenue related to the sales of the target product.

[0122] In some instances, some search terms are not applicable to the target product but, if applicable to the target product, may increase revenue. These currently inapplicable search terms may be referred to, in the context of advertisement, as “unattainable.” These unattainable search terms may be those search terms that are irrelevant, at least initially, to the target product for some reason or not yet associated with the target product because platform data associated with the digital marketplace **882** lacks data associated with the target product. In some embodiments, the machine learning module **896** may also be trained and used to receive data related to the characteristics of the target product, current competitors of the target product, and the current state of the ecommerce search term algorithm to determine the “winnability” of a search term. The winnability of a search term may be defined as the probability of winning each search term (e.g., having the target product associated with the search term) at any given point in time along with the estimated costs to win those search terms.

[0123] The machine learning module **896** may be trained with winnability inputs as described herein in order to provide a winnability report **804**. Some of the inputs for this model included any number of inputs and the description of certain types of inputs is not meant to limit the breadth of input into the machine learning module **896** in order to obtain a winnability report and the present specification contemplates these additional and different inputs. By way of example, an input may include a current and historical price for both the target product and competitive products. This historical pricing may be retrieved from one or more digital marketplaces **882** via the execution of the processor **810** and NID **880** as described herein. In this specific example, the processor **810** may cause the NID **880** to access the one or more digital marketplaces **882** either via a wired (wired transmitter/receiver **840**) or wireless (wireless transmitter/receiver **850**) connection, find instances of the target product and competing products being sold, and retrieve their historic

pricing values.

[0124] Another input to the machine learning module **896** may include a current and historical review ratings and review counts associated with the target product and competing products. These review ratings and review counts data may be retrieved from one or more digital marketplaces **882** via the execution of the processor **810** and NID **880** as described herein. Digital marketplaces **882** often provide a GUI that allows the consumer of the target product and competing products to enter text descriptive of the consumers' experiences with the target product and competing products as well as a ranked evaluation of those products in the form of a number rating system or star rating system. In this specific example, the processor **810** may cause the NID **880** to access the one or more digital marketplaces **882** either via a wired or wireless connection and find review ratings and review counts associated with the target product and competing products being sold, and provide that review ratings and review counts data to the machine learning module **896**.

[0125] Yet another input to the machine learning module **896** may include content similarity scores of any a search term related to the target product and competing products. These scores may be generated based on the data provided, in some embodiments. In FIG. **9**., for example, the further to the right any given search term is on the graph of FIG. **9**, the more similar the search terms of a competitive product are similar to the search terms of the target product. In a specific example, the x-axis (bottom) of the graph of FIG. **9**, or its associated data, may be used to assign this similarity score. As is shown in FIG. **9**, the similarity score may be either a positive or a negative score per the number ranking on the x-axis of FIG. **9**. In this example, the similarity score may be a positive weight or a negative weight reflected in the winnability report **804** provided by the processor **810** upon execution of the machine learning module **896**. In this embodiment, the processor **810** may, again, cause the NID **880** to access the one or more digital marketplaces **882** either via a wired or wireless connection and retrieve the plurality of sets of target product search terms and rankings as well as a plurality of sets of competing product search terms and rankings associated with the target product and competing products being sold. This data is then provided to the machine learning module **896**.

[0126] Still further, other input to the machine learning module **896** may include platform specific information such as average best seller rank (BSR) for any given digital marketplaces **882** associated with the target product and any number of competing products. A BSR may vary at any given digital marketplace **882**, but these rankings may be averaged over a plurality of digital marketplaces **882** to get this value. In this embodiment, the processor **810** may, again, cause the NID **880** to access the one or more digital marketplaces **882** either via a wired or wireless connection and retrieve this BSR data. This data is then provided to the machine learning module **896**.

[0127] Other input to the machine learning module **896** may include a projected search term volume and click distribution. In connection with this type of data provided to the machine learning module **896**, the projected search term volume may be retrieved from the data used to create the graph in FIG. **10**. This data describing how often any given search term associate with the target product and competing product appears in searches may be accessed by the processor **810** and provided as input to the machine learning module **896**. Additionally, any click distribution describing how many clicks any given search term gets may be accessed by the processor **810** and NID **880** at the search engine **894** of the digital marketplaces **882**.

[0128] Yet other input to the machine learning module **896** may include historical variations in search term ranks related to the target product and search phrase products. At any given time, a search engine **894** may have varying fluctuations in what is searched for on the internet. These search terms may be ranked and their historic ranking may change over time based on a number of social, political, environmental, and economic factors. This historical data may be retrieved from the search engine **894** by the processor **810** and NID **880** and provided to the machine learning module **896**.

[0129] Another example input to the machine learning module **896** may include targeted advertising spending associated with the search terms associated with the target product. This data may be maintained on any database that is accessible to the processor **810** of the computing device **822**. In a specific embodiment, this data descriptive of the targeted advertising spending associated with the search terms associated with the target product may be maintained by the seller of the targeted product on a private database and the user of the computing device **822** may be given secure access to that database. This type of data too may be provided to the machine learning module **896**.

[0130] With all of these different types of data obtained by the processor **810** via the NID **880**, the machine learning module **896** may build a number of mathematical models that provide a winnability report **804** that describes a probability of winning each search term (e.g., having the target product associated with the search term) at any given point in time along with the estimated costs to win those search terms. As with each machine learning module **896**, the machine learning module **896** may be “taught” by using the winnability factors described herein. In some embodiments, the machine learning module **896** may implement one or more non-parametric and parametric learning techniques. For example, the machine learning module, in some embodiments, may model the relationships between each plurality of sets of winnability factors using a layered neural network topology. Such a neural network, in some embodiments, may include an input layer (e.g., the winnability factors) including a known, recorded set of values for each of these parameters, settings, indicators, and usage data metrics, and an output layer including a projected winnability report **804**, based on the known, recorded set of values in the input layer. The machine learning module **896**, in some embodiments, may propagate input through the layers of the neural network to project or predict an optimal winnabilities of search terms based on the known and recorded search term metrics, and compare these projected values to optimal search terms to be presented in the winnability report **804**. Using a back-propagation method, the machine learning module **896**, in some embodiments, may then use the difference between the projected values and the known optimal values to adjust weight matrices of the neural network describing the ways in which changes in each of the search term data metrics are likely to affect the optimal search terms to be presented in the winnability report **804**.

[0131] With the output layer, the computing device **822** may provide learned competitive search terms that are determined to be the optimal search terms if any have been designated and based upon the winnable search terms detected at the search engine **894** of the digital marketplace **882** or other database during use of the computing device **822**. These resulting learned optimal search terms may be suggested to a user or automatically implemented. Suggestion may come with an indicator and may be shown in a graph at a user interface for, in some embodiments, approval by the user before implementation of the other processes executed by the processor **810** of the computing device **822**.

[0132] In some embodiments, the machine learning module **896** may perform a forward propagation and backward propagation, using different input node values repeatedly to finely tune any matrices either weighted or not. In such a way, the machine learning module **896**, in some embodiments, may adaptively learn how changes in the winnability factors may affect the data reflected in the winnability report **804**. The weight matrices associated with the layers of the neural network model in such an embodiment may describe, mathematically, these correlations for an individual target product. The neural network model (including designation of the node values in the input layer, and number of layers), along with the weight matrices associated with each layer, in some embodiments, may form a trained machine learning classifier, algorithm, or mathematical model to be used in generating any winnability report **804** as described herein.

[0133] As described herein, the output from the, now trained, machine learning module **896** is a winnability report **804**. With the winnability report **804** the computing device **822** may, with the processor **810** and NID **880**, determine a probability of attaining the desired change in revenue

based on a required investment. In some embodiments, the required investment may be calculated by the following equation:

$$\text{Required Investment} = \text{Projected Bid} * (\text{Impressions} * \text{Clickthrough Rate}) \quad \text{Equation 2}$$

[0134] A return on investment (ROI) may then be calculated using the following equation:

$$\text{ROI} = \text{Increased Revenue} * (\text{Projected Time to Remain at Required Investment}) \quad \text{Equation 3}$$

[0135] With Equations 2 and 3 those target products with search terms with high returns on investment can then be prioritized for both advertising and search engine optimization actions by the user. In this manner, the computing device **822** may execute the machine learning module **896** for a second purpose of determining the “winnability” of a search term where additional funds are applied to advertisements and search engine optimization.

[0136] In some embodiments, the ad spend margin, ad spend potential and revenue potential calculations by the processor **810** may also be conducted to specifically determine how much additional advertising funds to apply to the target product. Again, the ad spend margin may be calculated by multiplying a target ACoS by the price of the target product. A target ACOS may be determined and set by the seller based on available capital or may be set by the seller based on the fraction of the revenue received thus far from the sale of the target product on the digital marketplace **382** and costs of manufacturing. Ad spend potential may then be calculated by multiplying monthly opportunity units (OU) by the spend margin. The monthly OUs may be calculated as a result of the conversion rate of clicks to the target product that is the result of sales of the target product after a purchaser has viewed the product. The revenue potential may then be calculated by multiplying the OU with the price of the target product. This revenue potential of each of the target products may be ranked to determine the placement of the target product within the digital marketplace **882**. The search terms presented in the winnability report **804** may be sorted by revenue potential to determine the target product's best opportunities for revenue growth. In order to refine a recommendation, the process may continue with inputting estimated bid amounts from the digital marketplaces **882** required to win advertising slots for these keywords. In this manner, the execution of the processor **810** may initiate these calculations in order to predict a number of clicks and a cost necessary to achieve the potential growth. The equation to make this calculation is found in connection with Equation 2 herein.

[0137] An ROI may further be calculated by the following equation:

$$\text{ROI} = \text{Ad Spend Potential} * (\text{Investment Payoff Term} - \text{Investment Needed}) \quad \text{Equation 4}$$

[0138] As highly winnable terms are targeted in this process with both advertising and search engine optimization techniques, increasing the associated impressions, clicks, and conversions, the processing applied to the target product may continually adapt. As a target product succeeds on new search terms the competitive products set defined in the competitive set report **898** will shift to be compared to larger and less niche competing products. As the competitive products set defined in the competitive set report **898** shifts, the competitive terms set will shift as well. As reviews, terms, seller ranks, and other attributes shift, the winnability and associated required investment of each term also shifts. With the shift in winnability, new terms are prioritized and the cycle continues iteratively to cause the revenue associated with the targeted product to increase proportionally.

[0139] In some embodiments, the computing device **882** (e.g., and/or the system **100**, the computing device **420**, the computing device **520**, and/or the computing device **322**) may be configured to generate an optimal cost per click real-time bidding system balancing target spend and efficiency. For example, the computing device **882** may, for bidding campaigns that may otherwise run out of budget or overshoot desired spend, lower bids. Additionally, or alternatively, the computing device **882** may, for campaigns that underspend but have room given efficiency

targets, increase bids.

[0140] The computing device **882** may adjust the bids to achieve a return on advertising spend over a period of time. Campaigns that run out of budget or exceed spending targets, before a period ends, may not realize all potential revenue due to spending more than necessary over an undesirably short period of time. Alternatively, campaigns that do not spend enough budget, sacrifice sales volume. As such, the computing device **882** may provide automated computations that adjust bids to maximize efficiency while maintaining target spend. The computing device **882** may use one or more critical models and/or data processes to automatically adjust bids. The one or more critical models and/or data processes may include and/or use efficiency targets, estimated conversion rates, estimated order values, diminishing returns on incremental spend, and/or other suitable rates, values, data, and/or the like.

[0141] In some embodiments, the computing device **882** uses one or more of: a minimum efficiency and strategic requirements provided by a user; and desired spend within strategic requirements and minimum efficiency levels provided by the user. The computing device **882** may generate one or more models for individual targets including: a model to estimate conversion rate; a model to estimate order values; a model to estimate cost per click; and/or one or more other suitable models.

[0142] The computing device **882** may, using output from the one or more models, generate another model to predict tradeoff between bid, cost per click, and revenue to generate incremental revenue and incremental cost tradeoff. The computing device **882** may use a data stream to ingest data and predict, using the models, target bidding on a near real-time basis. The computing device **882** may receive user preferences, constraints, and a spending target, and may optimally adjust bids according to model outputs to achieve the desired spend, such that the desired spend is achievable given the constraints.

[0143] In some embodiments, the computing device **882** may receive user input data. In some embodiments, the user input data includes a desired total bid value and a bid period. In some embodiments, the user input data includes information associated with at least one product. In some embodiments, the user input data includes information associated with the at least one marketplace.

[0144] The computing device **882** may generate a first machine learning model that generates a predicted expected performance based on the user input data. The first machine learning model may include a tree-based machine learning model, a Bayesian machine learning model, or other suitable machine learning model.

[0145] The computing device **882** may adjust at least one bid on at least one of at least one keyword and at least one product associated with at least one marketplace, based on the predicted expected performance.

[0146] In some embodiments, the computing device **882** may generate a second machine learning model that generates a predicted expected cost based on the user input. The second machine learning model may include a tree-based machine learning model, a Bayesian machine learning model, or other suitable machine learning model. In some embodiments, the computing device **882** may adjust the at least one bid on the at least one of at least one keyword and the at least one product associated with the at least one marketplace, further based on the predicted expected cost.

[0147] FIG. **12** is a flow diagram illustrating a real-time bidding method **1200**, according to the principles of the present disclosure. At **1202**, the method **1200** receives user input data.

[0148] At **1204**, the method **1200** generates a first machine learning model that generates a predicted expected performance based on the user input data.

[0149] At **1206**, the method **1200** adjusts at least one bid on at least one of at least one keyword and at least one product associated with at least one marketplace, based on the predicted expected performance.

[0150] In some embodiments, a real-time bidding system includes a processor, and a memory. The memory includes instructions that, when executed by the processor, cause the processor to: receive

user input data; generate a first machine learning model that generates a predicted expected performance based on the user input data; and adjust at least one bid on at least one of at least one keyword and at least one product associated with at least one marketplace, based on the predicted expected performance.

[0151] In some embodiments, the user input data includes a desired total bid value and a bid period. In some embodiments, the user input data includes information associated the at least one product. In some embodiments, the user input includes information associated with the at least one marketplace. In some embodiments, the first machine learning model includes a tree-based machine learning model. In some embodiments, the first machine learning model includes a Bayesian machine learning model. In some embodiments, the instructions further cause the processor to generate a second machine learning model that generates a predicted expected cost based on the user input. In some embodiments, the second machine learning model includes a tree-based machine learning model. In some embodiments, the second machine learning model includes a Bayesian machine learning model. In some embodiments, the instructions further cause the processor to adjust the at least one bid on the at least one of at least one keyword and the at least one product associated with the at least one marketplace, further based on the predicted expected cost.

[0152] In some embodiments, a real-time bidding method includes receiving user input data, generating a first machine learning model that generates a predicted expected performance based on the user input data, and adjusting at least one bid on at least one of at least one keyword and at least one product associated with at least one marketplace, based on the predicted expected performance.

[0153] In some embodiments, the user input data includes a desired total bid value and a bid period. In some embodiments, the user input data includes information associated the at least one product. In some embodiments, the user input includes information associated with the at least one marketplace. In some embodiments, the first machine learning model includes a tree-based machine learning model. In some embodiments, the first machine learning model includes a Bayesian machine learning model. In some embodiments, the method also includes generating a second machine learning model that generates a predicted expected cost based on the user input. In some embodiments, the second machine learning model includes a tree-based machine learning model. In some embodiments, the second machine learning model includes a Bayesian machine learning model. In some embodiments, the method also includes adjusting the at least one bid on the at least one of at least one keyword and the at least one product associated with the at least one marketplace, further based on the predicted expected cost.

[0154] Any methods disclosed herein comprise one or more steps or actions for performing the described method. The method steps and/or actions may be interchanged with one another. In other words, unless a specific order of steps or actions is required for proper operation of the embodiment, the order and/or use of specific steps and/or actions may be modified.

[0155] Recitation in the claims of the term “first” with respect to a feature or element does not necessarily imply the existence of a second or additional such feature or element. Elements recited in means-plus-function format are intended to be construed in accordance with 35 U.S.C. § 112 Para. 6. It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the disclosure.

[0156] The above discussion is meant to be illustrative of the principles and various embodiments of the present disclosure. Numerous variations and modifications will become apparent to those skilled in the art once the above disclosure is fully appreciated. It is intended that the following claims be interpreted to embrace all such variations and modifications.

[0157] The word “example” is used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “example” is not necessarily to be construed as preferred

or advantageous over other aspects or designs. Rather, use of the word “example” is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from context, “X includes A or B” is intended to mean any of the natural inclusive permutations. That is, if X includes A; X includes B; or X includes both A and B, then “X includes A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form. Moreover, use of the term “an implementation” or “one implementation” throughout is not intended to mean the same embodiment or implementation unless described as such.

[0158] Implementations of the systems, algorithms, methods, instructions, etc., described herein can be realized in hardware, software, or any combination thereof. The hardware can include, for example, computers, intellectual property (IP) cores, application-specific integrated circuits (ASICs), programmable logic arrays, optical processors, programmable logic controllers, microcode, microcontrollers, servers, microprocessors, digital signal processors, or any other suitable circuit. In the claims, the term “processor” should be understood as encompassing any of the foregoing hardware, either singly or in combination. The terms “signal” and “data” are used interchangeably.

[0159] As used herein, the term module can include a packaged functional hardware unit designed for use with other components, a set of instructions executable by a controller (e.g., a processor executing software or firmware), processing circuitry configured to perform a particular function, and a self-contained hardware or software component that interfaces with a larger system. For example, a module can include an application specific integrated circuit (ASIC), a Field Programmable Gate Array (FPGA), a circuit, digital logic circuit, an analog circuit, a combination of discrete circuits, gates, and other types of hardware or combination thereof. In other embodiments, a module can include memory that stores instructions executable by a controller to implement a feature of the module.

[0160] Further, in one aspect, for example, systems described herein can be implemented using a general-purpose computer or general-purpose processor with a computer program that, when executed, carries out any of the respective methods, algorithms, and/or instructions described herein. In addition, or alternatively, for example, a special purpose computer/processor can be utilized which can contain other hardware for carrying out any of the methods, algorithms, or instructions described herein.

[0161] Further, all or a portion of implementations of the present disclosure can take the form of a computer program product accessible from, for example, a computer-usable or computer-readable medium. A computer-usable or computer-readable medium can be any device that can, for example, tangibly contain, store, communicate, or transport the program for use by or in connection with any processor. The medium can be, for example, an electronic, magnetic, optical, electromagnetic, or a semiconductor device. Other suitable mediums are also available.

[0162] The above-described embodiments, implementations, and aspects have been described in order to allow easy understanding of the present disclosure and do not limit the present disclosure. On the contrary, the disclosure is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation to encompass all such modifications and equivalent structure as is permitted under the law.

Claims

1. A real-time bidding system comprising: a processor; and a memory including instructions that, when executed by the processor, cause the processor to: receive user input data; generate a first

machine learning model that generates a predicted expected performance based on the user input data; and adjust at least one bid on at least one of at least one keyword and at least one product associated with at least one marketplace, based on the predicted expected performance.

2. The system of claim 1, wherein the user input data includes a desired total bid value and a bid period.

3. The system of claim 1, wherein the user input data includes information associated the at least one product.

4. The system of claim 1, wherein the user input includes information associated with the at least one marketplace.

5. The system of claim 1, wherein the first machine learning model includes a tree-based machine learning model.

6. The system of claim 1, wherein the first machine learning model includes a Bayesian machine learning model.

7. The system of claim 1, wherein the instructions further cause the processor to generate a second machine learning model that generates a predicted expected cost based on the user input.

8. The system of claim 7, wherein the second machine learning model includes a tree-based machine learning model.

9. The system of claim 7, wherein the second machine learning model includes a Bayesian machine learning model.

10. The system of claim 7, wherein the instructions further cause the processor to adjust the at least one bid on the at least one of at least one keyword and the at least one product associated with the at least one marketplace, further based on the predicted expected cost.

11. A real-time bidding method comprising: receiving user input data; generating a first machine learning model that generates a predicted expected performance based on the user input data; and adjusting at least one bid on at least one of at least one keyword and at least one product associated with at least one marketplace, based on the predicted expected performance.

12. The method of claim 11, wherein the user input data includes a desired total bid value and a bid period.

13. The method of claim 11, wherein the user input data includes information associated the at least one product.

14. The method of claim 11, wherein the user input includes information associated with the at least one marketplace.

15. The method of claim 11, wherein the first machine learning model includes a tree-based machine learning model.

16. The method of claim 11, wherein the first machine learning model includes a Bayesian machine learning model.

17. The method of claim 11, further comprising generating a second machine learning model that generates a predicted expected cost based on the user input.

18. The method of claim 17, wherein the second machine learning model includes a tree-based machine learning model.

19. The method of claim 17, wherein the second machine learning model includes a Bayesian machine learning model.

20. The method of claim 17, further comprising adjusting the at least one bid on the at least one of at least one keyword and the at least one product associated with the at least one marketplace, further based on the predicted expected cost.
